Physics Sample Exam Paper 3 . 45mins, 35Marks

វិញ្ញាសាប្រលងឆ្នាំ ២០១៦

I-Questions

1.What is focal length of optical lens?

2.What is mechanical advantage of lever?

II-Exercises

1.We want to lift a 200N object using a 1.5m long metal rod to be a lever. The effort is applied at other end.Calculate the effort force if the fulcrum is at 30cm from load.

2.A 1000W electric heater is used to boil 2kg of water from 20degrees until boil.

Assume that there is no energy lost and specific capacity of water is 4200J/kg^{0}C.

A.Calculate heat energy absorbed by water

B. Calculate the time taken to boil water

C. Calculate the cost per day if we boil 4times a day and the price per 1kWh is 720Riels.

3.A 20cm high object is placed at 40cm in front of a converging lens. The image appears at 10cm behind the lens. Calculate the focal length and image size.

Good Luck

Physics Sample Exam Paper 2

Exam 2017 (វគ្គកែ)

I. Questions

1.What is lens? How many types of lens are there? What are they?

2.What is lever? How many classes of lever? What are they?

II. Exercises

1.A 12cm high object is placed at 30cm in front of a convex lens whose focal length is 20cm.

A. Calculate the image location

B. Calculate the image size

2.An electric lamp having a power of 88W is connected a voltage of 220V.

A. Find the electric current flowing through the lamp.

B. Find the resistance of the lamp

c. The lamp is used 7hours per day, calculate the cost of using this lamp for 30days if 1kWh costs 720riels.

3. A transformer has 600turns at secondary coil and 200turns at the primary coil.

A. Is it the step up or step-down transformer?

B. Find the primary voltage if the secondary voltage is 240V.

C. Find the current in secondary coil if the primary current is 2A.

Good Luck

Physics Sample Exam Paper 4

I.Choose the correct answers

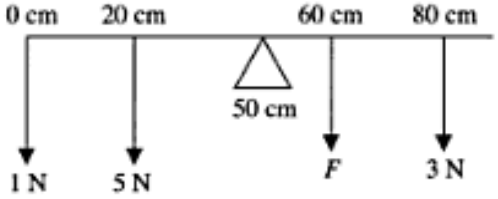
1.(2Marks)Mechanical advantage of a machine is defined as:

A. $MA = \frac{Effort}{Load}$ B. $MA = \frac{Force}{Load}$ C. $MA = \frac{Weight}{Load}$ D. $MA = \frac{Load}{Effort}$

2. (2Marks)A body can have:

A. two centers of gravity B. 3 centers of gravity
C. more centers of gravity D. Only one center of gravity

3. (2Marks)A long uniform beam is being balanced as shown Calculate the force F.



A. $F = 2N$ B. $F = 3N$ C. $F = 7N$ D. $F = 11N$

4. (2Marks) When you stand $5m$ in front of a plane mirror, your image is located,

a. $5m$ in front of the mirror. b. $5m$ behind the mirror.
c. less than $5m$ behind the mirror. d. more than $5m$ behind the mirror.

II.Problems

1. (When an electric kettle is connected to a 220 V mains supply, a current of 4.0 A passes through it.

(a) (3Marks)When the kettle is operating normally, what current is flowing in

(i) the live wire, _____ A

(ii) the neutral wire, _____ A

(iii) the earth wire, _____ A

(b) (1Marks) Calculate the power supplied to the kettle.

(c) (1Marks)Suggest a rating that could be used with this kettle.
(Available: Fuses of 2 A, 5 A, 10 A, 13 A ratings)

(d) (1Marks) Calculate the energy in kWh consumed by the kettle when it is switched on for 10 hours.

(e) (1Marks)The metal casing of the electric kettle is connected to the earth. Give one reason for this.

2. (5 marks) An electric kettle is used to boil $2kg$ of water from $50°C$. The kettle has a power rating of $1kW$ and an efficiency of 80%. How long does the water take to be boiled, provided that the specific heat capacity of water is $4200J/kg.°C$?

3. (10marks) A transformer has 800 primary turns and 2000 secondary turns. If the primary voltage is
160 V, determine the secondary voltage assuming an ideal transformer.

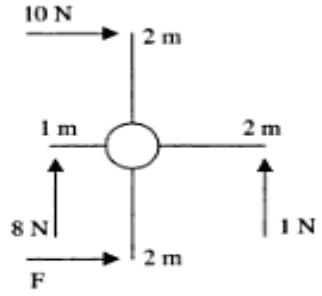
4. (5 marks) A concave lens has the focal lenth of 10cm and the object is palce at 12cm. Find the image distance.

Good Luck

Physics Sample Exam Paper 5

I.Choose the correct answers

1. (2 marks)A windmill is pushed by four external forces as shown. Calculate force F required to make the windmill in equilibrium.



A. $F=-10N$ B. $F=10N$

C. $F=13N$ D. $F=17N$

2. (2 marks) The figures below show the cross section of four solid objects. Which object is the most stable?



3. (2 marks) To determine the direction of the magnetic field produced by a straight current-carried wire, a right-hand rule is used where the direction the thumb points toward represents

a. the north pole. b. the direction of current flows.

c. the south pole. d. None of the above.

4. (2 marks) For concave curved mirror, the incident light that goes through the focal point will … off the mirror parallel to the principle axis.

a. refract b. reflect

c. incident d. diverge

II. (5marks) Match the column A with the column B

	A.Words		B.Meaning
1	Earth wire	A	It has the voltage about 220 Volts to your house and has the color of red or brown.
2	Live wire	B	A wire that is connected from the metal body of the appliances to Earth and, for normal function, it carries no electric current.
3	circuit breaker	C	a lever with wheel which can rotate about a fixed axis.

4	wheel and axle	D	An electromagnetic device that breaks a circuit when the current goes above a certain value.
5	gear	E	a wheel with teeth on its outer edge.

1	2	3	4	5

III. Problems

1. (4 marks) An electric kettle takes in 2000J of energy per second. It transfers 1500J as useful heat energy and 500J is wasted as sound energy. What is the efficiency of the kettle ?
2. (3 marks) A cylinder of a wheel and axle having a diameter of 40cm is used to lift a crate of 150N. The force applied to the handle is only 100N, Calculate the length of handle.
3. (8 marks) A water heater has a volume of 10litres and is fitted with a 2kW immersion heater. Calculate the time taken to raise the temperature of water from 40C to 740C.

   The water heater is 80 percent efficient. The specific heat capacity of water is 4200J/kg0C.
4. (2 marks) Calculate the speed of light in a plexiglass if the refractive index of plexiglass is 1.51.
5. (5 marks) A 2.25-cm-tall object is 8.5 cm to the left of a convex lens of 5.5-cm focal length. Find the image position and height.

……

Physics Sample Exam Paper 6

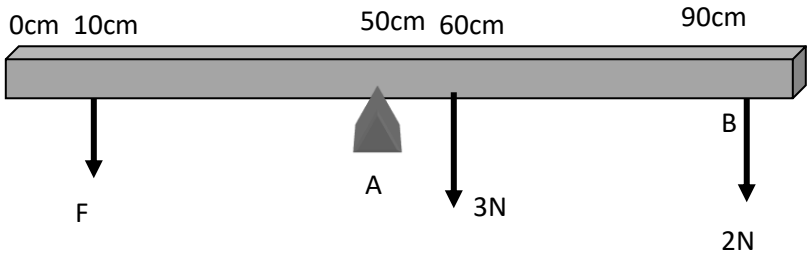
I.Choose the correct answers(8Marks)

1. A fan, with a resistance of 100Ω, is powered by a $240V$ socket. Suggest a suitable rating for a fuse to protect the fan from surge.

a. $1A$. b. $2A$.

c. $3A$. d. $5A$.

2- A uniform beam of 1m is being balanced as shown below. What is the moment of the 3N force about the pivot (point B).



A.10Ncm B.30Ncm C. 60Ncm D.90Ncm

3. For converging lenses (convex lenses), the incident light that is parallel to the principle axis will … through the focal point.

A. refract B. reflect

C. incident d. diverge

4. Which is the most appropriate fuse rating for a 240V, 1.1kW heater?

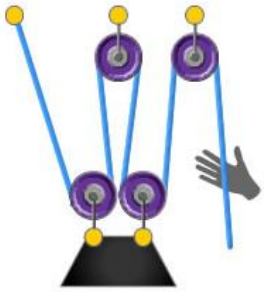
A. 1A B. 5A C. 13A D. 3A

II. Match the column A with the column B (4Marks)

A	B
1.Electric heater	a. has the highest potential
2. Live wire.	b. is any process in which electrical energy is converted to heat.
3. Optical lens	c. A device converts electric energy to heat energy
4. Electric heating	d. is an optical device that is used to bend light in a specific way.

III. Problems

1. (4 marks) For the pulley System below, find the MA of this pulley system.



2.(3 marks) A uniform meter stick weighs 1.4N, and is supported by a pivot at the 65cm mark. Where must the 0.85 N force be suspended to balance the stick?

3. (3 marks) A driver gear has 12 teeth and revolves at 100turns in 1minute. The driven gear has 36 teeth. What is the number of turn of the driven gear in 1 minute?

4.(5 marks) A transformer primary winding connected across a 415 V supply has 750 turns. Determine how many turns must be wound on the secondary side if an output of 1660V is required.

5. (5 marks) A 10-cm high object is placed 30 cm from a 20-cm focal length converging lens. Determine the image distance, the magnification of the image, the image height and the properties of the image.

Good Luck

Physics Sample Exam Paper 1

I. Choose the correct answers:

1.(2Marks) The principle of moment states that:

A. For a body to be in rotational equilibrium, the sum of clockwise moment must equal to the sum of anti-clockwise torques about the same point.

B. For a body to be in rotational equilibrium, the sum of clockwise force must equal to the sum of anti-clockwise torques about the same point.

C. For a body to be in rotational equilibrium, the sum of clockwise moment to the sum of anti-clockwise force about the same point.

D. For a body to be in rotational equilibrium, the sum of clockwise direction to the sum of anti-clockwise force about the same point.

2.(2Marks) Mechanical advantage:

A. is the number of load a machine multiplies the input (effort).

B. is the number of forces a machine multiplies the effort (input force).

C. is the number of times a machine multiplies the effort (input force).

D. is the number of efforts a machine multiplies the load (output force).

3.(2Marks) A transformer is a step down transformer when……

A. $\frac{N_s}{N_p} = 0$ B. $\frac{N_s}{N_p} > 1$

C. $\frac{N_s}{N_p} = 1$ D. $\frac{N_s}{N_p} < 1$

4.(2Marks) What is the element resistance of a heating rated as 1200W,240V?

A.5Ω B.48Ω C.12Ω D.120Ω

II. Problems

1.(5Marks) Calculate the resultant force of two forces(F1=4N, F2=12N) acting on a body. The two forces have the same original point and perpendicular to each other.

2.(5Marks) A tilted road is 50m long and 5m high above the ground Level. Calculate the work done and the force used to move a 100kg object to the top of road (Neglecting friction and g=10N/kg).

3.(5Marks) A 1000W electric heater is used to boil 2kg of water from 20degrees until boil. Assume that there is no energy lost and specific capacity of water is 4200J/kg^{0}C.

ក.Calculate heat energy absorbed by water.

ខ. Calculate the time taken to boil water

គ. Calculate the cost per day if we boil 4times a day and the price per 1kWh is 720Riels.

4. (5Marks) A transformer has 600turns at secondary coil and 200turns at the primary coil.

A. Is it the step up or step-down transformer?

B. Find the primary voltage if the secondary voltage is 240V.

C. Find the current in secondary coil if the primary current is 2A.

5.(7Marks) A 10cm high bright object is placed at 25cm in front of a convex lens who focal length 15cm. Determine the image distance.

Good Luck

Physics Sample Exam Paper 2

Exam 2017 (វិញ្ញកោះ)

I. Questions

1.What is lens? How many types of lens are there? What are they?

-A lens is an optical device that is used to bend light in a specific way.

-There are two types of thin lens: They are:

A **Converging Lens(Convex Lens)** is any lens that is thicker in the center than its edges.

A **diverging lens(Concave Lens)** is any lens that is thinner at the center than its edges.

2.What is lever? How many classes of lever? What are they?

A **lever** is a rigid rod, pivoted about a fixed point or axis, which is known as fulcrum.

There are 3 classes of lever, first class, second class, and third class.

II. Exercises

1.A 12cm high object is placed at 30cm in front of a convex lens whose focal length is 20cm.

A. Calculate the image location

Given: ho=12cm, do=30cm, f=20cm.

Using the equation: $\frac{1}{f}=\frac{1}{d_o}+\frac{1}{d_i}$ then we solve for di: $d_i=\frac{f\times d_0}{d_0-f}$

Replace the given value into the equation: $d_i=\frac{f\times d_0}{d_0-f}=\frac{20\times 30}{30-20}=60cm$

Therefore, the object distance is 60cm behind the lens.

B. Calculate the image size

$$M=\left|\frac{h_i}{h_o}\right|=\left|-\frac{d_i}{d_o}\right|\Rightarrow\left|\frac{h_i}{h_o}\right|=\left|-\frac{d_i}{d_o}\right|\Rightarrow hi=\left|\frac{ho\times di}{do}\right|=24cm$$

2.An electric lamp having a power of 88W is connected a voltage of 220V.

A. Find the electric current flowing through the lamp.

Given: P=88W, V=220V, find I.

Using the equation of electric power: $P=I\times V$

Solve for I: $\Rightarrow I=\frac{P}{V}=\frac{88}{220}=0.4A$

B. Find the resistance of the lamp

Using the Ohm's Law: $R = \frac{V}{I}$

Replace the given value into the equation: $R = \frac{V}{I} = \frac{220}{0.4} = 550\Omega$

c. The lamp is used 7hours per day, calculate the cost of using this lamp for 30days if 1kWh costs 720riels.

Given: t=7hoursx30days, 1kWh costs 720Riels, $E = P \times t = 0.088 \times 7 \times 30 = 18.48kWh$

Using $Cost = E(kWh) \times price\,per\,1kWh$

$Cost = 18.48 \times 720 = 13305.6 Riels/month$

3. A transformer has 600turns at secondary coil and 200turns at the primary coil.

A. Is it the step up or step-down transformer?

step-up transformer

B. Find the primary voltage if the secondary voltage is 240V.

Given: Vs=240V, Ns=600, Np=200

Using: $\frac{V_s}{V_p} = \frac{N_s}{N_p}$ Solve for Vp

So, $V_p = \frac{V_s \times N_p}{N_s} = \frac{240 \times 200}{600} = 80V$

C. Find the current in secondary coil if the primary current is 2A.

Using: $\frac{V_s}{V_p} = \frac{I_p}{I_s}$ Solve for Is

So, $I_s = \frac{V_p \times I_p}{V_s} = 0.67A$

Good Luck

Physics Sample Exam Paper 3

វិញ្ញាសាប្រលងឆ្នាំ ២០១៦

I-Questions

1.What is focal length of optical lens?

The distance from the vertical axis to the focal points

2.What is mechanical advantage of lever?

Mechanical advantage: is the number of times a machine multiplies the effort (input force).

II-Exercises

1.We want to lift a 200N object using a 1.5m long metal rod to be a lever. The effort is applied at other end.Calculate the effort force if the fulcrum is at 30cm from load.

Given: Fr =200N, dr =0.3m, de=1.5-0.3=1.2m.

$$MA=\frac{F_R}{F_E}=\frac{d_E}{d_R}\Rightarrow F_e=\frac{F_r\times d_r}{d_e}=\frac{200\times 0.3}{1.2}=50N$$

2.A 1000W electric heater is used to boil 2kg of water from 20degrees until boil.

Assume that there is no energy lost and specific capacity of water is 4200J/kg^{0}C.

A.Calculate heat energy absorbed by water

Given: P=1000W, m=2kg, $\Delta t=100-20=80$,c=4200J/kg^{0}C.

$$Q=m\times c\times \Delta\, t=2\times 4200\times 80=672000J$$

B. Calculate the time taken to boil water

For No energy lost: $Q=E=P\times t=672000$

Solve for t: $t=\frac{672000}{1000}=672s$

C. Calculate the cost per day if we boil **4**times a day and the price per 1kWh is 720Riels.

$$Cost=E\times 720=Riels/month$$

Given: P=1000W=1kW, $t=672s=0.187h$

$$Q=E=P\times t=0.187kWh$$

$$Cost=4\times E\times 720=.....Riels/month$$

3.A 20cm high object is placed at 40cm in front of a converging lens. The image appears at 10cm behind the lens. Calculate the focal length and image size.

Given: ho=20cm, do=40cm, di=10cm.

$$f=\frac{d_o\times d_i}{d_o+di}=\frac{40\times 10}{40+10}=8cm$$

$$M=\left|\frac{h_i}{h_o}\right|=\left|-\frac{d_i}{d_o}\right|\Rightarrow\left|\frac{h_i}{h_o}\right|=\left|-\frac{d_i}{d_o}\right|\Rightarrow hi=\left|\frac{ho\times di}{do}\right|=5cm$$

Physics Sample Exam Paper 4

I.Choose the correct answers

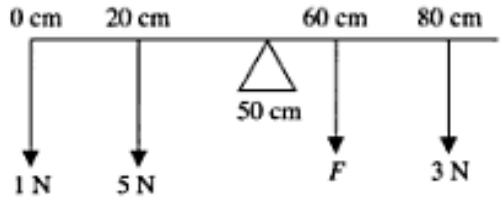
1.(2Marks)Mechanical advantage of a machine is defined as:

A. $MA=\frac{Effort}{Load}$ B. $MA=\frac{Force}{Load}$ C. $MA=\frac{Weight}{Load}$ D. $MA=\frac{Load}{Effort}$

2. (2Marks)A body can have:

A. two centers of gravity B. 3 centers of gravity
C. more centers of gravity D. Only one center of gravity

3. (2Marks)A long uniform beam is being balanced as shown Calculate the force F.



A. $F=2N$ B. $F=3N$ C. $F=7N$ D. $F=11N$

4. (2Marks) When you stand $5m$ in front of a plane mirror, your image is located,

a. $5m$ in front of the mirror. b. $5m$ behind the mirror.
c. less than $5m$ behind the mirror. d. more than $5m$ behind the mirror.

II.Problems

3. (When an electric kettle is connected to a 220 V mains supply, a current of 4.0 A passes through it.

(a) (3Marks)When the kettle is operating normally, what current is flowing in

(j) the live wire, _____ A

(ii) the neutral wire, _____ A

(iii) the earth wire, _____ A

(b) (1Marks) Calculate the power supplied to the kettle.

(c) (1Marks)Suggest a rating that could be used with this kettle.
(Available: Fuses of 2 A, 5 A, 10 A, 13 A ratings)

(d) (1Marks) Calculate the energy in kWh consumed by the kettle when it is switched on for 10 hours.

(e) (1Marks)The metal casing of the electric kettle is connected to the earth.

Give one reason for this.

4. (5 marks) An electric kettle is used to boil $2kg$ of water from $50°C$. The kettle has a power rating of $1kW$ and an efficiency of 80%. How long does the water take to be boiled, provided that the specific heat capacity of water is $4200J/kg.°C$?

Given: P=1000W, m=2kg, $\Delta t = 100-50 = 50$,c=4200J/kg^{0}C. $Rd = 80\% = 0.8$

$Q = m \times c \times \Delta t = 2 \times 4200 \times 50 = 420000J$, $E = P \times t = 1000t$

Using the efficiency: $Rd = \frac{Q}{E}$

Replace the given value into: $Rd = \frac{Q}{E} \Rightarrow 0.8 = \frac{420000}{1000t} \Rightarrow t = \frac{420000}{0.8 \times 1000} = 525s$

3. (10marks) A transformer has 800 primary turns and 2000 secondary turns. If the primary voltage is 160 V, determine the secondary voltage assuming an ideal transformer.

Given: Vs=240V, Ns=2000, Np=800,Vp=160V

Using: $\frac{V_s}{V_p} = \frac{N_s}{N_p}$ Solve for Vs

So, $V_s = \frac{V_p \times N_s}{N_p} = \frac{160 \times 2000}{800} = 400V$

4. (5 marks) A concave lens has the focal lenth of 10cm and the object is palce at 12cm. Find the image distance.

Given: do=12cm, f= -10cm.

Using the equation: $\frac{1}{f} = \frac{1}{d_o} + \frac{1}{d_i}$ then we solve for di: $d_i = \frac{f \times d_0}{d_0 - f}$

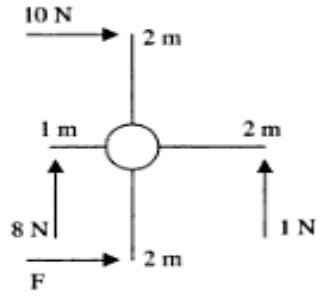
Replace the given value into the equation: $d_i = \frac{f \times d_0}{d_0 - f} = \frac{10 \times 12}{12 - (-10)} = ...cm$

Good Luck

Physics Sample Exam Paper 5

I.Choose the correct answers

1. (2 marks)A windmill is pushed by four external forces as shown. Calculate force F required to make the windmill in equilibrium.



A. $F=-10N$ B. $F=10N$

C. $F=13N$ D. $F=17N$

2. (2 marks) The figures below show the cross section of four solid objects. Which object is the most stable?



3. (2 marks) To determine the direction of the magnetic field produced by a straight current-carried wire, a right-hand rule is used where the direction the thumb points toward represents

a. the north pole. b. the direction of current flows.

c. the south pole. d. None of the above.

4. (2 marks) For concave curved mirror, the incident light that goes through the focal point will … off the mirror parallel to the principle axis.

a. refract b. reflect

c. incident d. diverge

II. (5marks) Match the column A with the column B

	A.Words		B.Meaning
1	Earth wire	A	It has the voltage about 220 Volts to your house and has the color of red or brown.
2	Live wire	B	A wire that is connected from the metal body of the appliances to Earth and, for normal function, it carries no electric current.

3	circuit breaker	C	a lever with wheel which can rotate about a fixed axis.
4	wheel and axle	D	An electromagnetic device that breaks a circuit when the current goes above a certain value.
5	gear	E	a wheel with teeth on its outer edge.

1	2	3	4	5

III. Problems

1. (4 marks) An electric kettle takes in 2000J of energy per second. It transfers 1500J as useful heat energy and 500J is wasted as sound energy. What is the efficiency of the kettle ?
2. (3 marks) A cylinder of a wheel and axle having a diameter of 40cm is used to lift a crate of 150N. The force applied to the handle is only 100N, Calculate the length of handle.
3. (8 marks) A water heater has a volume of 10litres and is fitted with a 2kW immersion heater. Calculate the time taken to raise the temperature of water from 40C to 740C.

   The water heater is 80 percent efficient. The specific heat capacity of water is 4200J/kg0C.
4. (2 marks) Calculate the speed of light in a plexiglass if the refractive index of plexiglass is 1.51.
5. (5 marks) A 2.25-cm-tall object is 8.5 cm to the left of a convex lens of 5.5-cm focal length. Find the image position and height.

Physics Sample Exam Paper 6

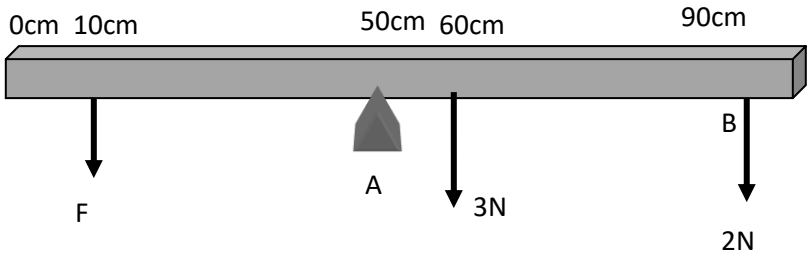
I.Choose the correct answers(8Marks)

1. A fan, with a resistance of 100Ω, is powered by a $240V$ socket. Suggest a suitable rating for a fuse to protect the fan from surge.

a. $1A$. b. $2A$.

c. $3A$. d. $5A$.

2- A uniform beam of 1m is being balanced as shown below. What is the moment of the 3N force about the pivot (point B).



A.10Ncm B.30Ncm C. 60Ncm D.90Ncm

3. For converging lenses (convex lenses), the incident light that is parallel to the principle axis will … through the focal point.

A. refract B. reflect

C. incident d. diverge

4. Which is the most appropriate fuse rating for a 240V, 1.1kW heater?

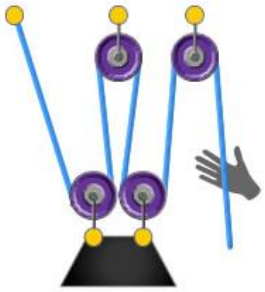
B. 1A B. 5A C. 13A D. 3A

II. Match the column A with the column B (4Marks)

A	B
1.Electric heater	a. has the highest potential
2. Live wire.	b. is any process in which electrical energy is converted to heat.
3. Optical lens	c. A device converts electric energy to heat energy
4. Electric heating	d. is an optical device that is used to bend light in a specific way.

III. Problems

2. (4 marks) For the pulley System below, find the MA of this pulley system.



2.(3 marks) A uniform meter stick weighs 1.4N, and is supported by a pivot at the 65cm mark. Where must the 0.85 N force be suspended to balance the stick?

3. (3 marks) A driver gear has 12 teeth and revolves at 100turns in 1minute. The driven gear has 37 teeth. What is the number of turn of the driven gear in 1 minute?

4.(5 marks) A transformer primary winding connected across a 415 V supply has 750 turns. Determine how many turns must be wound on the secondary side if an output of 1660V is required.

5. (5 marks) A 10-cm high object is placed 30 cm from a 20-cm focal length converging lens. Determine the image distance, the magnification of the image, the image height and the properties of the image.

Good Luck

Solution :

The image distance (d_i) :

$1/d_i = 1/f - 1/d_o = 1/20 - 1/30 = 3/60 - 2/60 = 1/60$

$d_i = 60/1 = 60$ cm

The plus sign indicates that the image is real or the rays pass through the image.

The magnification of image (m) :

$m = - d_i / d_o = -(60)/30 = -2$

The minus sign indicates that the image is inverted.

The image is 2 times greater than the object.

The image height (h_i) :

$m = h_i / h_o$

$hi = m\ h_o = (-2)10 = -20$ cm

The minus sign indicates that the image is inverted.

The properties of the image

– real

– inverted

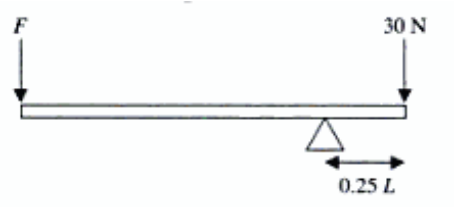
– the image height is greater than the object height

– the image distance is greater than the object distance

Physics Sample Exam Paper 7

I.Choose the correct answers

1- (3Marks)Two downward forces, F and 30N, act on a light uniform plank of length **L** at the two ends as shown. Calculate the force F to make the plank equilibrium.



A.10N B.30N C. 40N D.120N

2. (2Marks)When a light ray travels from air into water, it

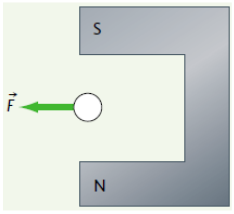
a. reflects and refracts. b. converges and diverges.

c. reflects only at the point of incident. d. refracts only.

3. (3Marks)Fill in the table below:

Output Force (N)	Input Force (N)	Mechanical Advantage
200		5
	100	4
500	250	

4. (2Marks) Choose the direction of current for the given magnetic force and Field below.



A. To the left B. into page C. Out of page D. upwards

II. Problems

1-(6Marks) A screw-type pump is used to raise 200 kg of water a distance of 3 m.

The handle of the pump is turned with an effort force of 40 N a total of 1200 times tracing a circle of 20 cm circumference each time.

(a) Calculate the work output.

(b) Calculate the work input.

(c) Calculate the efficiency.

2. (6Marks) A 6m light beam is supported at the ends A and B. The weight of 50kN is placed at 2.5m from A. Find the downward forces exert on the supports A and B.
3. (6Marks) Where must an object be placed to form an image 30 cm from a diverging lens with a focal length of 40 cm? Determine the magnification of the image.
4. (7Marks)A consumer requires an immersion heater for a tank containing 150liters of Water. The water is to be heated from 10^0C to 60^0C in 3hours. Calculate the element size (in kilowatts) if the heating system is 90 percent efficient. The specific heat capacity of water is 4200J/kg^{0}C.

Good Luck

Physics Sample Exam Paper 8

I.Choose the correct answers

1. Transformers are used to increase/decrease voltage. If we desire a secondary voltage to be smaller than a primary voltage, then
   a. the secondary coil needs to has less turns than the primary coils.
   b. the secondary coil needs to has more turns than the primary coils.
   c. the number of turns in both coils has to be equal.
   d. the ratio between the number of turns in both coils must be smaller than zero.

II. Problems

1.The convex lens of a copy machine has a focal length of 25.0 cm. A letter to be copied is placed 40.0 cm from the lens.

a. How far from the lens is the copy paper?

b. How much larger will the copy be?

2.

វិញ្ញាសាទី២ រូបវិទ្យា ថ្នាក់ទី (៩)

សម្រាប់រៀមប្រឡង (រយៈពេល ៤៥ នាទី)

I. **ចូរគូសសញ្ញា (√) ក្នុងប្រអប់ខាងមុខចម្លើយត្រឹមត្រូវ៖**

១. ម៉ូម៉ង់កម្លាំងជាទំហំមួយកំណត់ដោយផលគុណរវាងកម្លាំង និង

☐ ក. ប្រវែងប្តង់ទេ ☑ ខ. ប្រវែងដៃឃ្នាស ☐ គ. ប្រវែងខ្សែរែក ☐ ឃ. ប្រវែងកាំ

២. អង្គធាតុមួយមានទីប្រជុំទម្ងន់តែ

☑ ក. 1គត់ ☐ ខ. 2គត់ ☐ គ. 3គត់ ☐ ឃ. 4គត់

II. **ចូរបំពេញល្បះខាងក្រោមឲ្យបានត្រឹមត្រូវ**

១. ចម្ងាយរវាងអ័ក្សរង្វិល ទៅនឹងខ្សែសកម្មនៃកម្លាំងហៅថា រដ្ឋធានីភ្នំពេញ។

២. ទីប្រជុំទម្ងន់នៃផែនទីប្រទេសកម្ពុជាស្ថិតនៅក្នុងខេត្ត ក្រចេះ។

៣. អង្គធាតុមួយចល័តជុំវិញអ័ក្សដេក កាលណាទីប្រជុំទម្ងន់របស់វាស្ថិតនៅខាងក្រោមអ័ក្ស ដែលកាត់តាមខ្សែឈរ នៃអ័ក្សនោះគេថា អង្គធាតុនោះមាន សមតុល្យ។

៤. អង្គធាតុមួយដែលអាចចល័តជុំវិញអ័ក្សមួយមានលំនឹងកាលណា ផលបូកម៉ូម៉ង់កម្លាំងចាប់លើអង្គធាតុនោះ ....ស្មើសូន្យ....។

III. **ចូរផ្គូផ្គងឆ្នៃក (A) និង (B) ឲ្យបានត្រឹមត្រូវ**

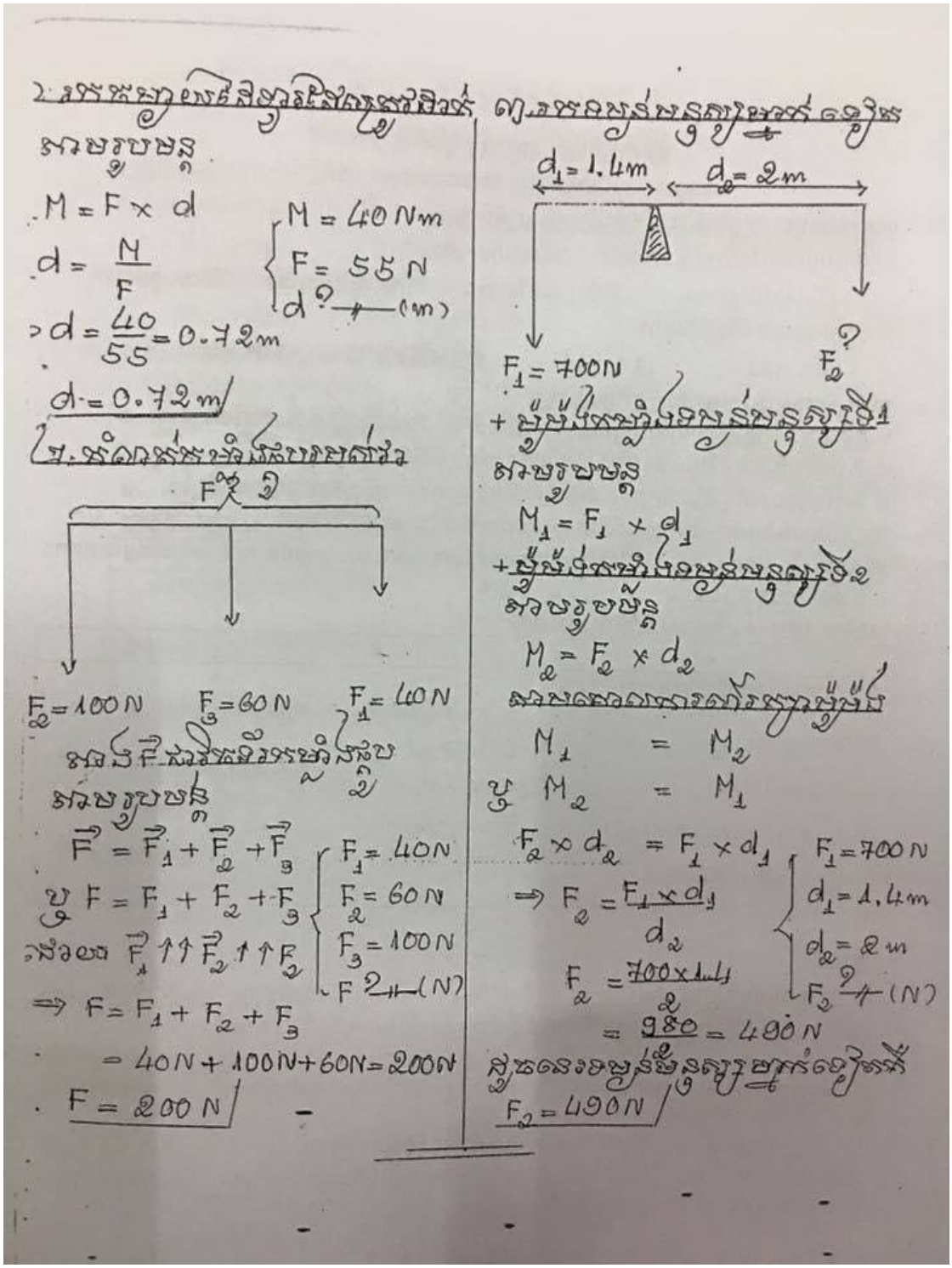
A	B	ចម្លើយ
១. រូបមន្តផលបូកកម្លាំងពីរស្របគ្នា ទិសដៅដូចគ្នា	ក. $\vec{F}=\vec{F}_1+\vec{F}_2$ ឬ $F=F_1+F_2$	១. → ង
២. រូបមន្តម៉ូម៉ង់កម្លាំង	ខ. $M=F\times d$	២. → ខ
៣. រូបមន្តម៉ូម៉ង់ទម្ងន់របស់មនុស្សទី1	គ. $M_1=M_2$	៣. → ឃ
៤. តាមគោលការណ៍រក្បាម៉ូម៉ង់	ឃ. $M_1=F_1\times d_1$	៤. → គ
	ង. $M_2=F_2\times d_2$	

IV. **លំហាត់**

១. ទ្វារមួយមានម៉ូម៉ង់អប្បបរមា $40Nm$ ដើម្បីបិទបើកទ្វារ។ តើគេត្រូវដាក់ដៃទ្វារនៅប្រវែងចម្ងាយប៉ុន្មាន (m) ពីត្រចៀក ? បើគេដឹងថា ដើម្បីបិទ ឬបើកទ្វារ គេត្រូវបង្ហោះកម្លាំងចាន $55N$ ។

២. បន្ទាត់ស្មើសាច់មួយមានប្រវែង $100cm$ រងទ្រកម្លាំងប៉ស្របគ្នា និងមានទិសដៅដូចគ្នា។ ចុងខាងស្តាំរងទ្រកម្លាំងមួយស្មើនឹង $40N$ និងចុងខាងឆ្វេងរងទ្រងកម្លាំងមួយស្មើនឹង $100N$ ហើយនៅកណ្តាលរងទ្រកម្លាំងស្មើនឹង $60N$ ។ កំណត់កម្លាំងផ្គួបរបស់វា ។

៣. មនុស្សម្នាក់មានទម្ងន់ $700N$ អង្គុយនៅចុងម្ខាងនៃដងខ្លឹងមួយចម្ងាយ $1.4m$ ពីអ័ក្សរង្វិល។ ដើម្បីឲ្យដងខ្លឹងមានលំនឹងស្ថិតតាមទិសដេក ។ តើគេត្រូវដាក់មនុស្សម្នាក់ទៀតមានទម្ងន់ប៉ុន្មាន (N) ? ដែលអង្គុយនៅចុងម្ខាងទៀតនៃដងខ្លឹង ដែលមានប្រវែង $2m$ ពីអ័ក្សរង្វិល ។


ក. រកចម្ងាយដែលកម្លាំងទីពីរ
ត្រូវដាក់
ដោយប្រើ
M = F × d
d = M / F
M = 40 Nm
F = 55 N
d ? (m)
d = 40/55 = 0.72 m
d = 0.72 m
ខ. គណនារង្វាស់កម្លាំង
F ?
F2 = 100 N
F3 = 60 N
F1 = 40 N
តម្លៃ F ដែលមានឥទ្ធិពលស្មើនឹង
ដោយប្រើ
F = F1 + F2 + F3
ឬ F = F1 + F2 + F3
ដែល F1 ↑↑ F2 ↑↑ F3
F1 = 40N
F2 = 60 N
F3 = 100 N
F ? (N)
⇒ F = F1 + F2 + F3
= 40N + 100N + 60N = 200N
F = 200 N
ឃ. រកចម្ងាយដែលកម្លាំងទីពីរ
d1 = 1.4m
d2 = 2m
F1 = 700N
F2 ?
+ ម៉ូម៉ង់កម្លាំងបង្កើតបង្វិលទី1
ដោយប្រើ
M1 = F1 × d1
+ ម៉ូម៉ង់កម្លាំងបង្កើតបង្វិលទី2
ដោយប្រើ
M2 = F2 × d2
តាមលក្ខណៈនៃតុល្យភាពម៉ូម៉ង់
M1 = M2
ឬ M2 = M1
F2 × d2 = F1 × d1
⇒ F2 = F1 × d1 / d2
F1 = 700 N
d1 = 1.4m
d2 = 2 m
F2 ? (N)
F2 = 700 × 1.4 / 2
= 980 / 2 = 490 N
ដូចនេះចម្ងាយដែលកម្លាំងទីពីរត្រូវដាក់
F2 = 490 N

ឥណ្ហាសាធីរ រូបវិទ្យា ថ្នាក់ទី (៩)
សម្រាប់ត្រៀមប្រឡង (រយៈពេល ៤៥ នាទី)

I. **ចុះក្នុងសញ្ញា (√) ក្នុងប្រអប់ខាងមុខចម្លើយត្រឹមត្រូវ៖**

១. ផលមេកានិចនៃប្រព័ន្ធកែស្មើនឹងចំនួនខ្សែដែល

☑ ក. ទ្របន្ទុក ☐ ខ. ទ្រឃ្នាស់ ☐ គ. ទ្រប្លង់ទេ ☐ ឃ. កម្លាំងចលករ

២. កែចល័តគឺជាកែដែលផ្លាស់ទីតាមទិសដៅ

☐ ក. កម្លាំងទប់ ☑ ខ. កម្លាំងចលករ ☐ គ. កម្លាំងទាញ ☐ ឃ. កម្លាំងរុញ

II. **ចុះបំពេញល្បះខាងក្រោមឲ្យបានត្រឹមត្រូវ**

១. បង្គុំកម្លាំងពីរស្រប‌គ្នាជាកម្លាំង.....ដិបផ្ទាល.....។

២.....ស្ថានភាពនៃបាល់ដែលកម្លាំងទប់ ឬបន្ទុកស្ថិតនៅចន្លោះកម្លាំងចលករ (F_E) និង ទម្រ (o)។

៣. ក្នុងការប្រើប្លង់ទេរកម្ពស់ដែលប្រវែងប្លង់ទេរកាន់តែវែងផលមេកានិច.....តិចជាង.....។

៤.....ស្ថិតក្នុងដៃ..... ជាទំហំមួយកំណត់ដោយផលគុណរវាងកម្លាំង និងប្រវែងដៃឃ្នាស់។

III. **ចុះផ្គូផ្គងថ្នែក (A) និង (B) ឲ្យបានត្រឹមត្រូវ**

A	B	ចម្លើយ
១. តាមច្បាប់ក្បោកម្មន្តនៃប្លង់ទេរ	ក. $MA=\frac{F_R}{F_E}$ ឬ $MA=\frac{L}{h}$	១. → ឲ
២. រូបមន្តកម្មន្តដើម្បីលើកវត្ថុត្រង់ទៅលើ	ខ. $W_E=W_R$	២. → ឃ
៣. រូបមន្តទម្ងន់ ឬកម្លាំង	គ. $p=F_R=mg$	៣. → ក
៤. រូបមន្តផលមេកានិចនៃប្លង់ទេរ	ឃ. $W_R=F_R\times h$	៤. → ង
	ង. $W_T=F_T\times L$	

IV. **លំហាត់**

១. បន្ទាត់ស្មើសាច់មួយមានប្រវែង $60cm$ ចុងទាំងសងខាងនៃបន្ទាត់រងទ្រកម្លាំងពីរស្រប‌គ្នា និងមានទិសដៅផ្ទុយ‌គ្នា។ កម្លាំងទីមួយស្មើនឹង $700N$ កម្លាំងទីពីរស្មើនឹង $500N$។ កំណត់កម្លាំងផ្ទួបរបស់វា។

២. គេមានកម្លាំងពីរ $\vec{F_1}$ និង $\vec{F_2}$ កែងគ្នាមានអាំងតង់ស៊ីតេរៀងគ្នា $6N$ និង $8N$ ហើយមានចំណុចចាប់ផ្តើម O។

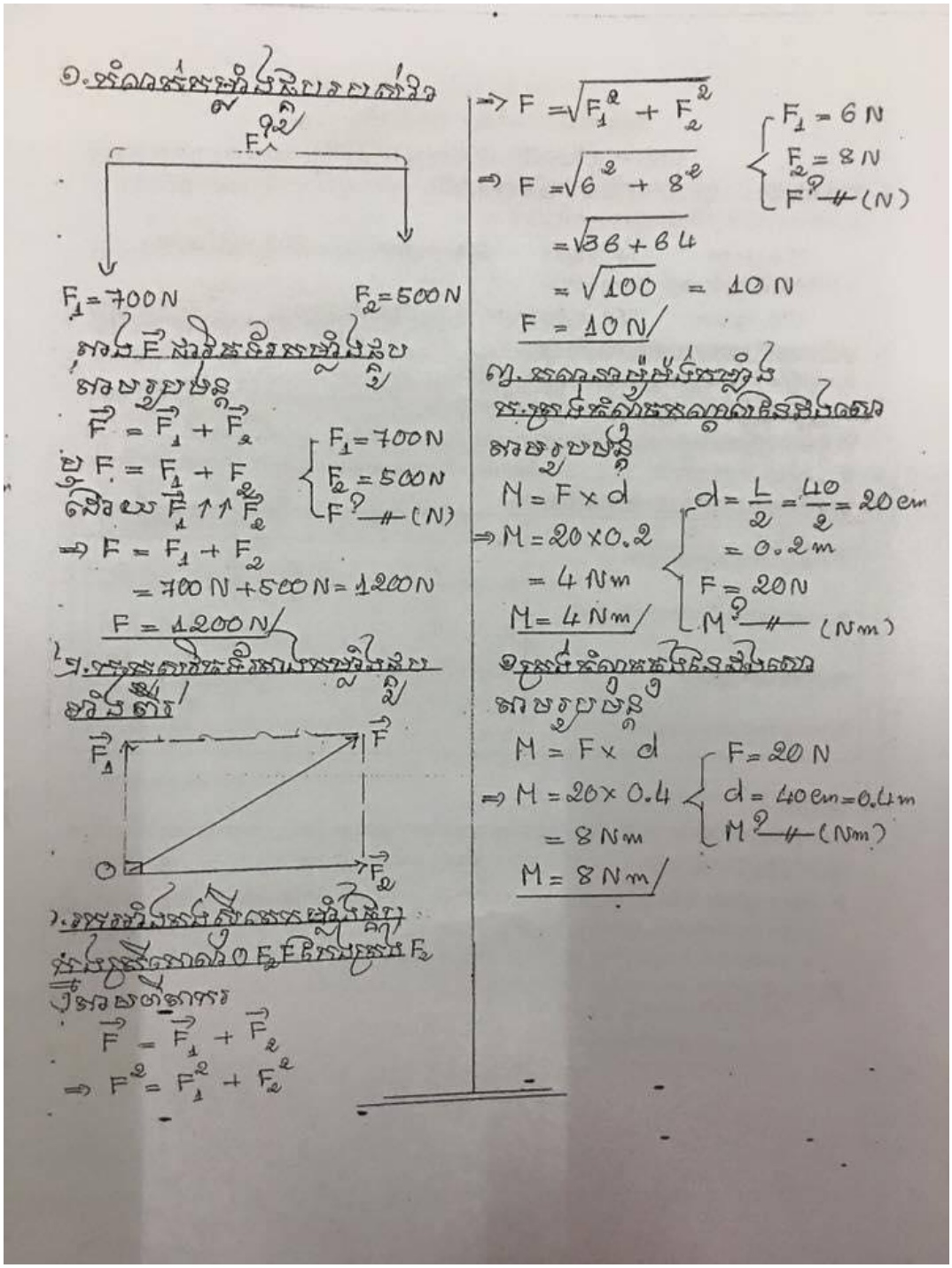
ក. ចូរកំណែទម្រង់កម្លាំងផ្ទួបទាំងពីរ (កម្លាំងផ្ទួប)របស់វា។

ខ. រកអាំងតង់ស៊ីតេកម្លាំងផ្ទួបដោយប្រើមាត្រដ្ឋាន $1cm=1N$។

៣. គេបញ្ចេញកម្លាំង $20N$ កែងនឹងសោមួយមានប្រវែង $40cm$។ គណនាម៉ូម៉ង់កម្លាំង:

ក. ត្រង់ចំណុចកណ្តាលនៃដងសោ ?

ខ. ត្រង់ចំណុចខាងចុងនៃដងសោ ?



ក. រកតម្លៃនៃកម្លាំងសម្មតិ

ដែល $\vec{F}_1 \uparrow\uparrow \vec{F}_2$

$\Rightarrow F = F_1 + F_2$

$= 700N + 500N = 1200N$

$F = 1200N$

ខ. រកតម្លៃនៃកម្លាំងសម្មតិ

ដែល $\vec{F}_1 \perp \vec{F}_2$

គ. រកទំហំនៃកម្លាំងសម្មតិ

ដែលមុំរវាង $\vec{F}_1$ និង $\vec{F}_2$

ជាមុំខាងស្តាំ

តាមទ្រឹស្តីបទ

$\vec{F} = \vec{F}_1 + \vec{F}_2$

$\Rightarrow F^2 = F_1^2 + F_2^2$

$\Rightarrow F = \sqrt{F_1^2 + F_2^2}$

$\Rightarrow F = \sqrt{6^2 + 8^2}$ $\begin{cases} F_1 = 6N \\ F_2 = 8N \\ F = ? \text{ (N)} \end{cases}$

$= \sqrt{36 + 64}$

$= \sqrt{100} = 10N$

$F = 10N$

ក. រកតម្លៃនៃម៉ូម៉ង់

ខ. ម៉ូម៉ង់គូកម្លាំងទី១

តាមរូបមន្ត

$M = F \times d$ $\begin{cases} d = \frac{L}{2} = \frac{40}{2} = 20cm \\ = 0.2m \\ F = 20N \\ M = ? \text{ (Nm)} \end{cases}$

$\Rightarrow M = 20 \times 0.2$

$= 4Nm$

$M = 4Nm$

ខ. ម៉ូម៉ង់គូកម្លាំងទី២

តាមរូបមន្ត

$M = F \times d$ $\begin{cases} F = 20N \\ d = 40cm = 0.4m \\ M = ? \text{ (Nm)} \end{cases}$

$\Rightarrow M = 20 \times 0.4$

$= 8Nm$

$M = 8Nm$

វិញ្ញាសាធី៤ រូបវិទ្យា ថ្នាក់ទី (៩)

សម្រាប់រៀមប្រឡង (រយៈពេល ៤៥ នាទី)

I. **ចូរគូសសញ្ញា (√) កុងប្រអប់ខាងមុខចម្លើយត្រឹមត្រូវ៖**

១. វ៉ែកនឹងគឺជាវ៉ែកដែលមិនផ្លាស់ប្តូរទីតាំងរបស់វានៅពេលគេលើកដាក់

☑ ក. បន្ទុក ☐ ខ. កម្លាំង ☐ គ. រយៈពេល ☐ ឃ. ចម្ងាយ

២. វ៉ែកគឺជាកង់ម្យ៉ាង ដែលធ្វើអំពីឈើ ឬដែក និងមាន

☐ ក. ព័ទ្ធជុំវិញត្រឹយ ☑ ខ. ចង្អូរព័ទ្ធជុំវិញ ☐ គ. ព័ទ្ធជុំវិញឃ្នាស់ ☐ ឃ. ព័ទ្ធជុំវិញប្លង់ទេ

II. **ចូរបំពេញល្បះខាងក្រោមឲ្យបានត្រឹមត្រូវ**

១. អង្គធាតុមួយ ឬវត្ថុមួយ មានទីប្រជុំទម្ងន់តែ.....មួយគត់.....។

២. ផលមេកានិចនៃប្រព័ន្ធវ៉ែកស្មើនឹងចំនួនខ្សែ.....ដឹងសាច់ខានេះ.....។

៣. ចម្ងាយរវាងអ័ក្សរង្វិល ទៅនឹងខ្សែសកម្មនៃកម្លាំងហៅថា.....ដឹងដែរដឹងខ្លួន.....។

៤. ក្នុងការប្រើវ៉ែកចល័តមួយ អាំងតង់ស៊ីតេកម្លាំងចលករស្មើពាក់កណ្តាលអាំងតង់ស៊ីតេកម្លាំងទប់ ឬ បន្ទុក (.....$F_E = F_R$.....)។

III. **ចូរផ្គូផ្គងផ្នែក (A) និង (B) ឲ្យបានត្រឹមត្រូវ**

A	B	ចម្លើយ
១. ផលមេកានិចនៃកង់យោង	ក. $F_R = P = mg$	
២. រូបមន្តទម្ងន់ ឬបន្ទុក	ខ. $M_A = \frac{F_E}{F_R}$ ឬ $M_A = \frac{R}{r}$	១. → ឯ
៣. តាមច្បាប់រក្សាកម្មន្ត	គ. $M_1 = M_2$	២. → ជ
៤. រូបមន្តចម្ងាយពីចំណុចចាប់នៃកម្លាំងផ្តុំ	ឃ. $\frac{d_2}{d_1} = \frac{F_1}{F_2}$	៣. → ក
ជាភាគច្រាសសមាមាត្រនិងអាំងតង់ស៊ីតេ	ង. $M = F \times d$	៤. → ខ

IV. **លំហាត់**

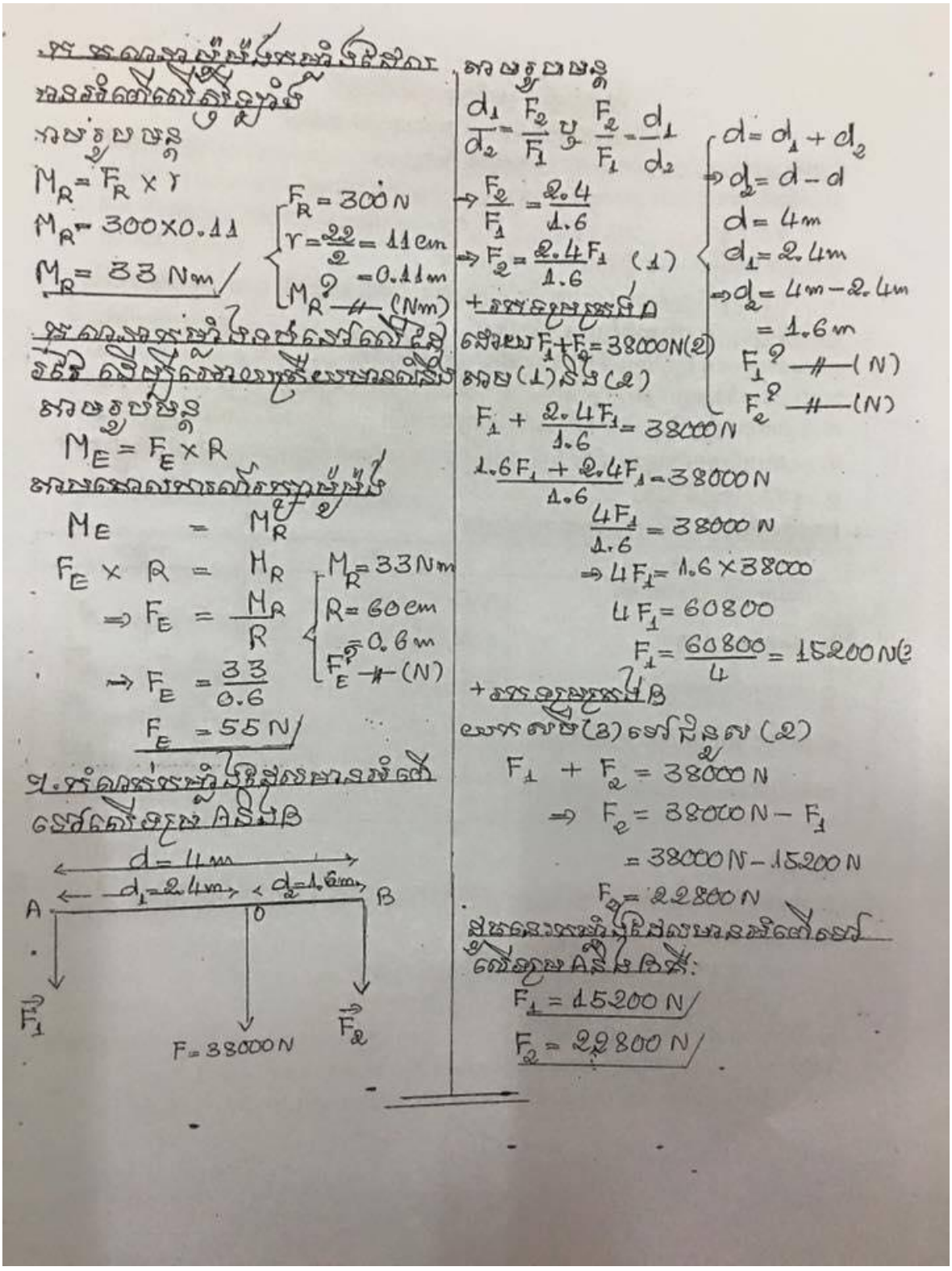
១. គេព្យួរបន្ទុកមួយមានទម្ងន់ $300N$ នៅលើអង្គត់ធ្ងិតនៃស៊ីឡាំងរបស់ត្រឹយមួយមានប្រវែង $22cm$ និងកាំនៃរង្វង់គូសដោយដៃវៃមានប្រវែង $60cm$ ។

ក. គណនាម៉ូម៉ង់នៃកម្លាំងដែលមានអំពើលើស៊ីឡាំង ?

ខ. គណនាកម្លាំងទប់នៅលើដៃវៃ ដើម្បីឲ្យត្រឹយមានលំនឹង ។

២. គេមានរបារដែកមួយទ្រដោយទម្រ A និង B ដែលមានចម្ងាយពីគ្នា $4m$ ។ របារនោះទ្រទម្ងន់មួយស្មើនឹង $P = 38000N$ ត្រង់ចំណុចចាប់មួយដែលមានចម្ងាយពី A ប្រវែង $2.4m$ ។

កំណត់កម្លាំងដែលមានអំពើទៅលើទម្រ A និង B (ទម្ងន់របារអាចចោលបាន)។



ប្រឡងសញ្ញាបត្រមធ្យមសិក្សាបឋមភូមិ ចំណេះទូទៅនិងបំពេញវិជ្ជា
សម័យប្រឡង: ថ្ងៃទី07 ខែកញ្ញាឆ្នាំ 2017

វិញ្ញាសា រូបវិទ្យា

រយៈពេល 45នាទី 35ពិន្ទុ

I-សំនួរ(6ពិន្ទុ)

1-តើគេត្រូវ តខ្សែណឹត ឬ ខ្សែជាស ចូលទៅក្នុងកុងតាក់? (2ពីន្ទុ)

3- ឧបករណ៍ដែលជាម៉ាស៊ីនងាយមាន អ្វីខ្លះ? (2ពីន្ទុ)

3-ត្រង់ស្វូមានប៉ុន្មានប្រភេទអ្វីខ្លះ? (2ពីន្ទុ)

II-លំហាត់(29ពិន្ទុ)

1-ប្លង់ទេរមួយមានរាងត្រីកោណប្រវែង 4mហើយបង្កើតបានជាមុំθ= 30° ។ គេត្រូវការរុញវត្ថុដែលមានទម្ងន់ 200N ។តើគេត្រូវប្រើកម្លាំងប៉ុន្មានដើម្បីរុញវត្ថុនោះឱ្យដល់ទិសដៅ? ($\sin 30^\circ = 0.5$) (6ពិន្ទុ)

②ឆ្នាំងអ៊ុតអគ្គិសនីមួយស៊ីអានុភាព1kw។ គេប្រើឆ្នាំងអ៊ុតនេះដើម្បីអ៊ុតខោ អាវអស់រយៈពេល 3/4នៃម៉ោង ។

ក- គណនាថាមពលអគ្គិសនី គិតជា (J) និង (KWh)។*(6ពិន្ទុ)*

ខ- បើគេប្រើឆ្នាំងអ៊ុតនេះក្នុងមួយថ្ងៃ 4ដង តើត្រូវចំណាយប្រាក់អស់ប៉ុន្មាន?បើថ្លៃអគ្គិសនី 2kwh ស្មើ 1440រ ។ *(5ពិន្ទុ)*

3-នៅពីមុខឡង់ទីបង្រួមដែលមានចម្ងាយកំណុំ 6cmមានវត្ថុភ្លឺ AB មួយ ។ កំណត់ទីវត្ថុនិងចម្ងាយរូបភាពក្នុងករណី:

ក- ពីវត្ថុពិត AB ឡង់ទីឱ្យរូបភាពពិតធំជាងវត្ថុ 2ដង ។ (6ពិន្ទុ)

ខ- ពីវត្ថុពិត AB ឡង់ទីឱ្យរូបភាពមិនពិតធំជាងវត្ថុ 2ដង ។ (6ពិន្ទុ)